Practice Problem Set 4

ECON 101 – Macroeconomics

Winter 2026

Part I: Chapter 6 — Economic Growth and the Financial System

1. Explain the difference between:

- economic growth,
- business cycles.

Provide one real-world example of each.

2. Why is **labour productivity** crucial for long-run economic growth? Explain using:

- capital deepening,
- technological progress.

3. Explain how **financial markets and financial intermediaries** contribute to economic growth. What would happen if financial markets were inefficient?

4. Loanable Funds Model

Suppose:

- National saving = \$800 billion,
- Investment demand = \$800 billion.

Now the government runs a deficit of \$200 billion.

(a) What happens to national saving?

(b) Show how the supply of loanable funds changes.

(c) What happens to the equilibrium interest rate?

(d) Explain crowding out.

5. **Growth Accounting**

   Suppose:

   - Real GDP grows by 4%,
   - Labour input grows by 1%.

   (a) Calculate labour productivity growth.

   (b) What factors could explain this growth?

6. Why do some developing countries fail to converge to the income levels of developed countries? Discuss at least three structural reasons.

Part II: Chapter 8 — Aggregate Expenditure and Output

7. Define:

   - Marginal Propensity to Consume (MPC),
   - Marginal Propensity to Save (MPS).

   Explain their relationship.

8. Why does the **multiplier effect** amplify changes in spending? Explain intuitively.

9. **Multiplier Calculation**

   Suppose:

   - MPC = 0.8,
   - Increase in government spending = $50 billion.

   (a) Calculate the multiplier.

   (b) Calculate the change in equilibrium GDP.

10. **Aggregate Expenditure Model**

    Suppose:

    $$C = 200 + 0.75Y, \quad I = 150, \quad G = 250$$

    (a) Write the aggregate expenditure (AE) equation.

    (b) Find equilibrium GDP.

    (c) Calculate the multiplier.

    (d) If G increases by 40, find the new equilibrium GDP.

11. **Recessionary Gap**

    - Potential GDP = 3,000,
    - Actual GDP = 2,600,
    - MPC = 0.8.

    (a) Calculate the GDP gap.
    (b) Calculate the required increase in government spending to close the gap.
    (c) Explain the policy intuition.

Part III: Chapter 9 — Aggregate Demand and Aggregate Supply

12. Explain why the aggregate demand (AD) curve is downward sloping using:

    - wealth effect,
    - interest rate effect,
    - international trade effect.

13. Distinguish between:

    - short-run aggregate supply (SRAS),
    - long-run aggregate supply (LRAS).

14. What causes shifts in:

    - AD,
    - SRAS,
    - LRAS?

    Give one example for each.

15. **Demand Shock**

    Suppose consumer confidence falls.

    (a) What happens to AD?
    (b) What are the short-run effects on:
        - output,
        - price level,
        - unemployment?

16. Supply Shock

Suppose oil prices increase significantly.

(a) What happens to SRAS?

(b) What happens to inflation and output?

(c) What type of inflation is this?

17. Suppose:

- Potential GDP = 2,500,
- Actual GDP = 2,700,
- Inflation is rising.

(a) Is there an inflationary or recessionary gap?

(b) What policy should be used?

(c) Explain the adjustment process back to equilibrium.

18. The economy experiences:

- a decline in investment,
- rising interest rates,
- falling GDP.

(a) Explain using:
  - AE model,
  - AD–AS model.

(b) What role does the financial system play in this downturn?

(c) Propose one fiscal policy and one monetary policy. Explain how each affects GDP.

19. Suppose:

- Government spending increases by $100,
- MPC = 0.75,
- Interest rates rise due to deficit financing.

Explain why the actual increase in GDP may be smaller than predicted by the multiplier.

End of Problem Set